

CSC4093 / DSC4213: Neural Networks and Deep Learning

Programming Assignment 01

PHM Tweet Classification using LSTM and Bi-LSTM

S20344

a. Performance Comparison

Two models were built and tested on the PHM tweet dataset (9,991 training tweets, 3,331 test tweets). The dataset has a class imbalance of 2.45:1 (Non-PHM to PHM).

Figure 1 below shows the class distribution across the training and test sets.

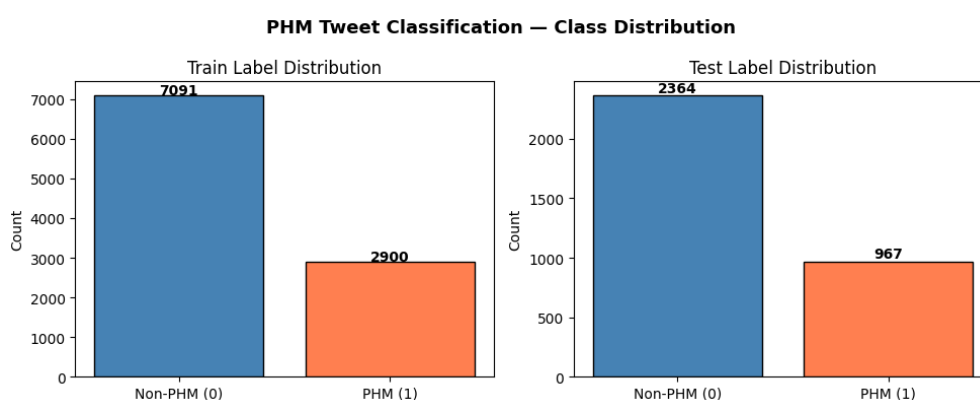


Figure 1: PHM Tweet Classification — Class Distribution (Train & Test)

Interpretation:

The dataset is clearly imbalanced, with more Non-PHM tweets than PHM tweets. This imbalance can bias the model toward predicting the majority class, which is why class weighting and appropriate evaluation metrics are necessary.

Table 1: Performance Comparison of LSTM and Bi-LSTM on Test Set

Metric	LSTM (t=0.45)	Bi-LSTM (t=0.49)	Better Model
Accuracy	0.8343	0.8235	LSTM
Precision	0.7155	0.6773	LSTM
Recall	0.7125	0.7487	Bi-LSTM
Weighted F1	0.8342	0.8260	LSTM
Macro F1	0.7987	0.7920	LSTM
ROC-AUC	0.8917	0.8953	Bi-LSTM

Interpretation:

The LSTM achieved 83.4% accuracy and the Bi-LSTM achieved 82.4% accuracy. The LSTM wins on accuracy, precision, weighted F1, and macro F1. The Bi-LSTM has better recall (0.749 vs 0.713), meaning it catches more actual health tweets, and a slightly better ROC-AUC (0.895 vs 0.892).

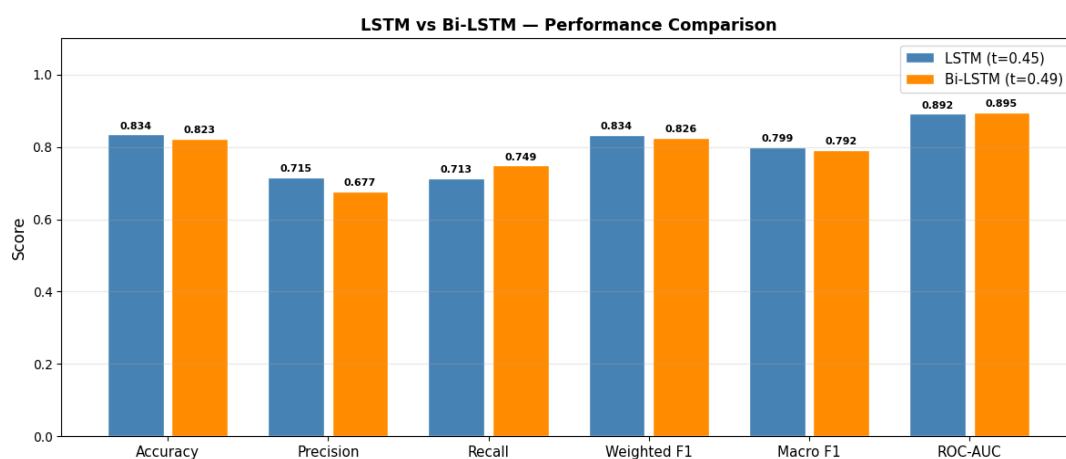


Figure 2: LSTM vs Bi-LSTM — Performance Metrics Bar Chart

Interpretation:

The LSTM model performs slightly better on most metrics such as accuracy, precision, and F1-scores. However, the Bi-LSTM achieves higher recall and ROC-AUC, indicating better ability to detect PHM tweets.

5-fold cross-validation was also done to check if the results are consistent. **Table 2** shows the average results across 5 folds.

Table 2: 5-Fold Cross-Validation Results (mean $\pm$ std)

Metric	LSTM (mean $\pm$ std)	Bi-LSTM (mean $\pm$ std)
Accuracy	0.8077 $\pm$ 0.0093	0.8102 $\pm$ 0.0081
Macro F1	0.7773 $\pm$ 0.0072	0.7828 $\pm$ 0.0059
ROC-AUC	0.8748 $\pm$ 0.0069	0.8815 $\pm$ 0.0048

Interpretation:

The cross-validation shows both models perform very similarly (about 81% accuracy). The small standard deviations (less than 1%) confirm that both models are stable and generalize well to unseen data.

Training History

Figures 3 and 4 show the accuracy and loss curves for the LSTM and Bi-LSTM models during training. Figure 5 overlays both models for direct comparison.



Figure 3: LSTM — Training and Validation Accuracy & Loss

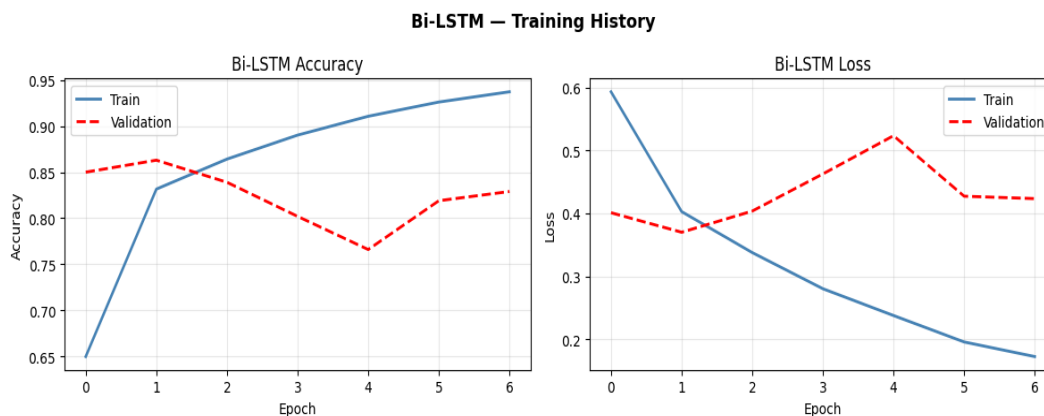


Figure 4: Bi-LSTM — Training and Validation Accuracy & Loss

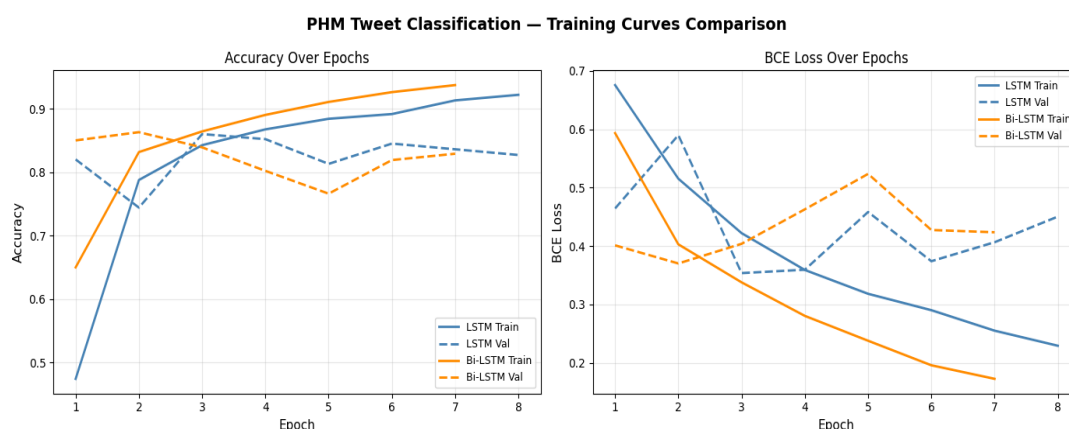


Figure 5: PHM Tweet Classification — Training Curves Comparison (LSTM vs Bi-LSTM)

Interpretation:

The Bi-LSTM begins to overfit earlier than the LSTM, as shown by increasing validation loss after fewer epochs. The LSTM maintains better balance between training and validation performance, explaining its superior test results.

Confusion Matrices

Figure 6 shows the confusion matrices for both models on the test set with their respective optimised thresholds.

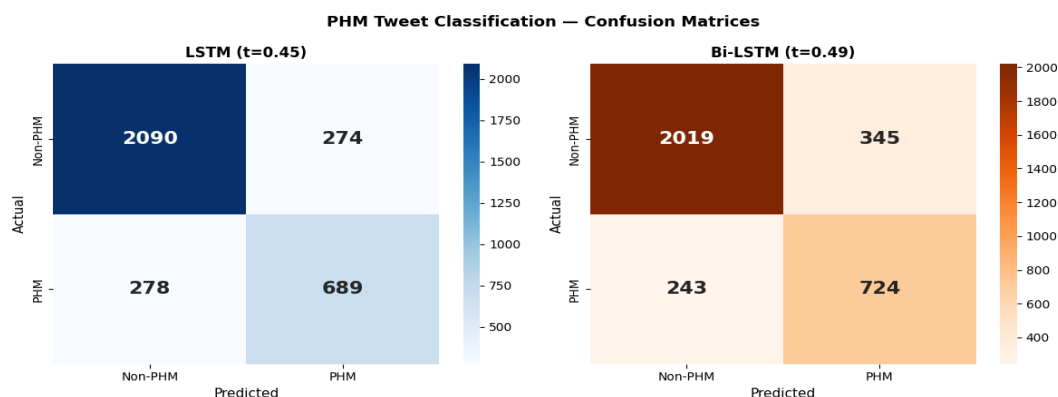


Figure 6: PHM Tweet Classification — Confusion Matrices (LSTM $t=0.45$ | Bi-LSTM $t=0.49$)

Interpretation:

The Bi-LSTM correctly identifies more PHM tweets, confirming its higher recall. However, this comes at the cost of slightly more false positives.

ROC and Precision-Recall Curves

Figure 7 shows the ROC curves and Precision-Recall curves for both models. The stars (★) mark the chosen classification threshold for each model.

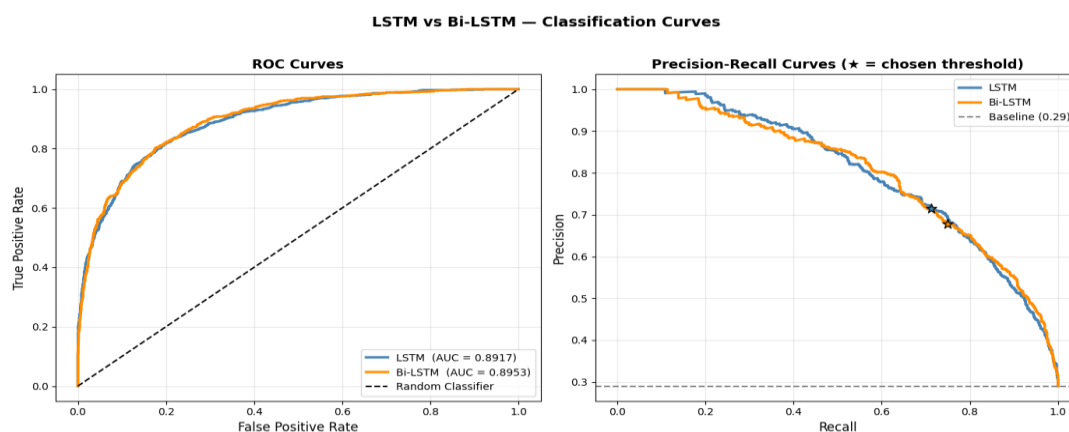


Figure 7: LSTM vs Bi-LSTM — ROC Curves and Precision-Recall Curves

Interpretation:

Both models show strong discriminative ability, with curves well above the random baseline. The chosen thresholds improve Macro F1 score instead of relying on the default 0.5 cutoff.

b. Discussion on the Performance of the Two Models

Model Architecture

Both models use the same structure: an Embedding layer (64 dimensions), SpatialDropout1D (0.3), an LSTM or Bi-LSTM layer (64 units), a Dense layer (32 units with L2 regularisation), Dropout (0.3), and a Sigmoid output. The only difference is that the Bi-LSTM reads tweets in both forward and backward directions, which doubles its number of parameters in the recurrent layer (128-dimensional output vs 64).

Why LSTM Beat Bi-LSTM

Normally, Bi-LSTM performs better than LSTM on text tasks because it can use context from both directions. However, in this case the LSTM did slightly better. The main reasons are:

- Tweets are short (≈ 11 words), so backward context adds limited value.
- Bi-LSTM has more parameters and overfits faster.
- LSTM generalizes better under limited context.

Techniques Used to Improve Performance

- Custom stopword list preserving pronouns and negations
- Class weighting to address imbalance
- Threshold optimization
- Dropout and L2 regularization
- Early stopping and learning-rate reduction

Conclusion

Both LSTM and Bi-LSTM models perform well for PHM tweet classification. The LSTM achieves the best overall balance across accuracy and F1-scores, while the Bi-LSTM provides higher recall and ROC-AUC. Cross-validation confirms consistent and stable performance for both models. Given the short length of tweets, the simpler LSTM architecture proves more effective and less prone to overfitting.